Lab Worksheet 1
Basic Configuration and Set-up of Routers and Switches

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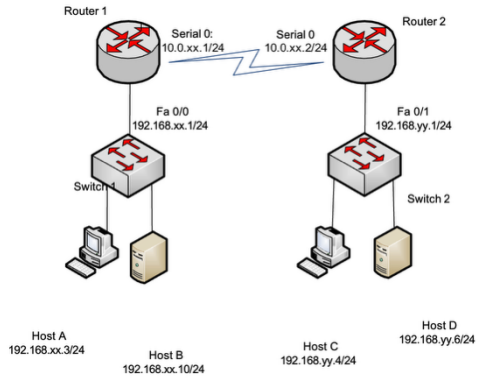
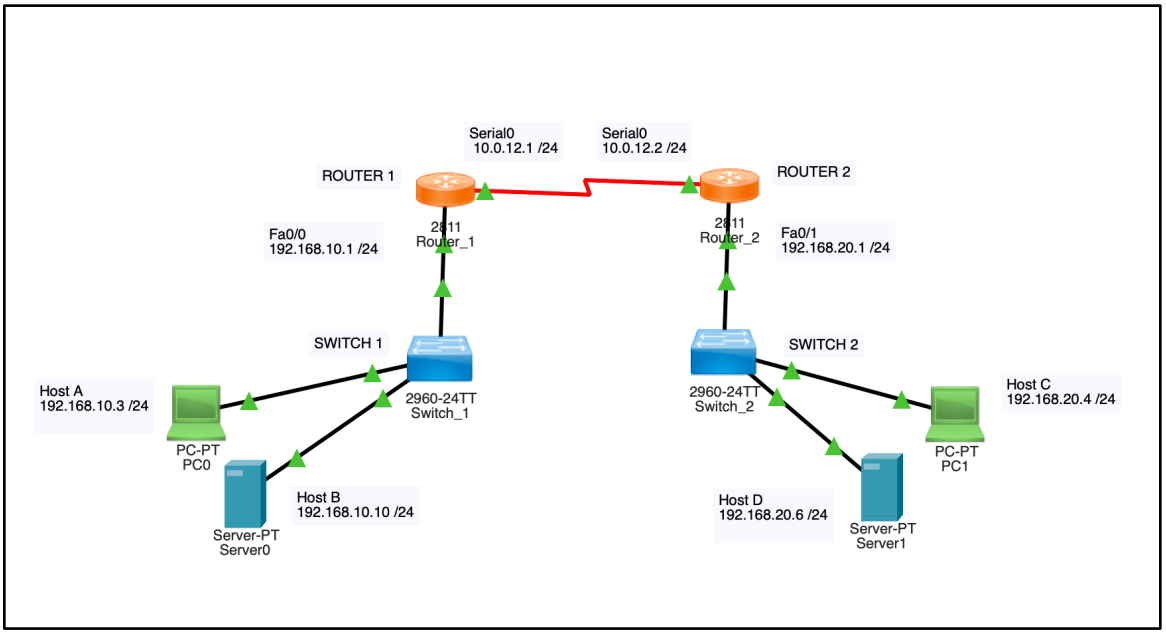
Lab Exercise



Task 1

Set-up the above network topology and save it as *Lab1 Basic Configurations Pair xx Block y*. (Example Lab 1 Basic Configuration Pair 01 Block 1)



Be mindful of the cable being used to connect the devices.

Task 2.

Configure the routers and switches with details provided on the above topology.
Named the routers as:
Router 1 → Router_1
Router 2 → Router_2

Task 3.

Step 1: Perform basic connectivity test from hosts to the routers.

a. Is ping successful from

Host A to Host B? **YES**

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time=1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time=9ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 2ms

C:\>
```

Host A to Router 1? **YES**

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Host A to Router 2's serial 0? **YES**

```
C:\>ping 10.0.12.2

Pinging 10.0.12.2 with 32 bytes of data:

Reply from 10.0.12.2: bytes=32 time=4ms TTL=254
Reply from 10.0.12.2: bytes=32 time=14ms TTL=254
Reply from 10.0.12.2: bytes=32 time=13ms TTL=254
Reply from 10.0.12.2: bytes=32 time=14ms TTL=254

Ping statistics for 10.0.12.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 14ms, Average = 11ms

C:\>
```

Router 1 to Router 2? **YES**

```
Router_1>enable
Password:
Router_1#ping 10.0.12.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.12.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 15/19/23 ms

Router_1#
```

Host A to Host C? **YES**

```
C:\>ping 192.168.20.4

Pinging 192.168.20.4 with 32 bytes of data:

Reply from 192.168.20.4: bytes=32 time=3ms TTL=126
Reply from 192.168.20.4: bytes=32 time=1ms TTL=126
Reply from 192.168.20.4: bytes=32 time=15ms TTL=126
Reply from 192.168.20.4: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.20.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 15ms, Average = 5ms

C:\>
```

Host C to Host D? **YES**

```
C:\>ping 192.168.20.6

Pinging 192.168.20.6 with 32 bytes of data:

Reply from 192.168.20.6: bytes=32 time=1ms TTL=128
Reply from 192.168.20.6: bytes=32 time<1ms TTL=128
Reply from 192.168.20.6: bytes=32 time<1ms TTL=128
Reply from 192.168.20.6: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

b. Perform a trace from Host A to Host D, what is your observation on the result? Use the **tracert** command.

The tracert output shows successful end-to-end connectivity from **Host A (192.168.10.3)** to **Host D (192.168.20.6). Zero packet loss and all devices are responding.**

```
C:\>tracert 192.168.20.6

Tracing route to 192.168.20.6 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.10.1
  2   0 ms      1 ms      5 ms      10.0.12.2
  3   0 ms      5 ms      0 ms      192.168.20.6

Trace complete.

C:\>
```

Step 2: Telnet either from Host A to Router 1 or Host C to Router 2. What is your observation on the result? Use the **telnet** command.

The Telnet session was **completely successful**. I connected to Router 2 (from Host C) via its Serial0 interface (10.0.12.2), authenticated with the VTY password, entered privileged mode using the enable password, and even accessed global configuration mode before exiting cleanly.
Without setting passwords and enabling login on the VTY lines, Telnet is impossible. The router will **refuse the connection entirely** (if VTY lines aren't configured at all)

```
C:\>telnet 10.0.12.2
Trying 10.0.12.2 ...Open


User Access Verification

Password:
Password:
Router_2>enable
Password:
Router_2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router_2(config)#exit
Router_2#exit

[Connection to 10.0.12.2 closed by foreign host]
C:\>
```

Task 4.
Using **show** commands, verify the following;

1. **show version**
IOS version of Router 1:
- **Version 15.1(4)m4**

IOS image of Router 1:
- "**flash0:c2800nm-advipservicesk9-mz.151-4.M4.bin"**

uptime of Switch 2:
- **39 minutes**

RAM of Router 2:
- **cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory**

Configuration register of Router 2:
- **Configuration register is 0x2102**

2. **show interface Serial 0**

Interface status in Router 1:
- **Serial0/2/0 is up, line protocol is up (connected)**

Bandwidth:
- **BW 1544 Kbit**

Reliability:
- **reliability 255/255**

MTU:
- **MTU 1500**

bytes

Output drops:

- **Total output drops: 0**

Duplex:

- **Full-Duplex**

3. **show ip interface brief**

Describe the output of Router 2:

The show ip interface brief command on **Router 2** displays the status of all its interfaces. **FastEthernet0/1** is configured with IP address **192.168.20.1 and is up/up**, indicating it is actively connected to the right-side **(192.168.20.0/24)** and serving as the default gateway for **Hosts C and D**.

```
Router_2>enable
Password:
Router_2#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES unset  administratively down down
FastEthernet0/1        192.168.20.1    YES manual up                    up
Serial0/2/0            10.0.12.2       YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router_2#
```

4. **show cdp neighbor**
Describe the output in Router 1

The show cdp neighbors output on Router 1 reveals two directly connected devices: **Router_2** connected via **Serial0/2/0**, confirming the connection is operational, and **Switch_1** connected via **FastEthernet0/0**, confirming the connection to the switch.

```
Router_1#show cdp neighbor
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID    Local Intrfce   Holdtme    Capability   Platform    Port ID
Router_2     Ser 0/2/0        161            R       C2800       Ser 0/2/0
Switch_1     Fas 0/0          170            S       2960        Fas 0/3
Router_1#
```

5. **show run**
Describe the output in Router 2

The running configuration of Router 2 shows two active interfaces: **FastEthernet0/1** is configured with IP address **192.168.20.1/24** for the right-side LAN, and **Serial0/2/0** is configured with IP address **10.0.12.2/24** for the WAN link to Router 1. Both interfaces are operational and properly configured. The remaining interfaces—**FastEthernet0/0** and **Vlan1**—are **shutdown** and not in use.

```
interface FastEthernet0/0
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface FastEthernet0/1
 ip address 192.168.20.1 255.255.255.0
 duplex auto
 speed auto
!
interface Serial0/2/0
 ip address 10.0.12.2 255.255.255.0
!
interface Vlan1
 no ip address
 shutdown
!
ip classless

Router_2#
```

II. Conclusion

In this activity, I successfully set up a small network topology consisting of two routers, two switches, and four host devices. I learned that proper IP addressing is essential for communication between devices, and that serial interfaces require additional modules (WIC-1T) to establish a link between routers. I also discovered that CDP (Cisco Discovery Protocol) is a useful tool for verifying directly connected neighbors and confirming network topology. Most importantly, I learned that configuring administrative settings such as VTY passwords and enable secrets is necessary to allow remote access via Telnet — without these, the router will reject connection attempts. Through commands like ping, tracert, telnet, show ip interface brief, show version, and show running-config, I was able to verify connectivity, troubleshoot issues, and understand the role of each device in the network. This activity reinforced the importance of both physical connectivity and logical configuration in building a functional and manageable network.

End of Lab Worksheet 1